Worksheet 1

Assignment 3

CS 2470 Questions

Question 1

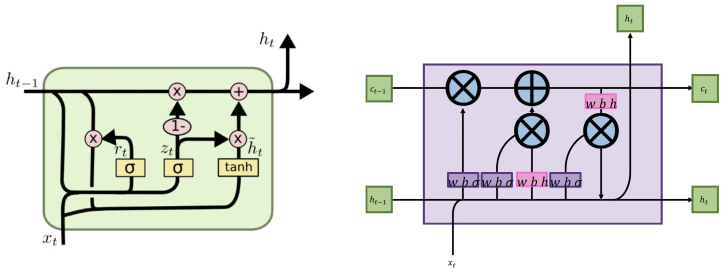


Figure 1: lstm vs gru

Compared with LSTM cell, GRU cell has less tensor computation. GRU cells have mainly three types of gates: reset gate, update gate and output gate. The update gate is similar to forget and input gate of an LSTM: it decides the deletion and addition of new information at the current cell. The reset gate is another gate used to decide how much past information to forget. The output gate can give the new cell state. In GRU, the update and reset gate together can decide what information can be forgot and what information can be retained, for both short and long term memory.

Question 2

The paper I looked up is:
Bai S, Kolter JZ, Koltun V. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv preprint arXiv:1803.01271. 2018 Mar 4.
In consensus, practioners commonly take recurrent neural networks as the default starting point for sequence modeling tasks. The pros for this model is that it has the ability to pass long or short term memory, but on the other hand, it is notoriously hard for training such model.
Convolutional neural networks(CNN) were predominately used in areas such as computer vision, image recognition, or image segmentation. The convolution layer is specially designed for capturing local features of the images. However, more than a decade ago, researches has already been starting to use CNN in sequential modeling such as speech recognition, sentence classification. From the results, the CNN model exhibits a great advantage in a comprehensive suite of sequence modeling tasks, with an averaged testing accuracy at 99%,

where the accuracies for LSTM and GRU are around 20% and 15%. However, it is still too early to assert that CNN is a way better model in sequence modeling.